

LESSON GUIDE

Technology Plus

Unit 7: Security

6.2.3 – Device Use Best Practices

Lesson Overview:

This lesson examines the best practices for securing devices through proper software use, emphasizing the importance of legitimate licensing, identifying trusted sources, removing unnecessary or malicious programs, and understanding the implications of software piracy.

Students will:

- Differentiate between open-source and proprietary licensing models.
- Compare subscription-based, one-time purchase, and perpetual licensing models.
- Explain the importance of product keys and serial numbers.
- Identify trustworthy software sources and explain how to verify them.
- Describe when and why software should be removed.
- Analyze the ethical and legal consequences of software piracy.

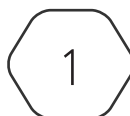
Guiding Question: How can we make smart software choices that protect devices and maintain legality and security?

Suggested Grade Levels: 8-12

Technology Needed: Yes

Approximate Time Required: 60-75 minutes

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Lesson Background

Standards:

CompTIA Tech+ FC0-U71 Objective

- 6.2 Explain methods to secure devices and security best practices.
 - Device use best practices
 - Licensing
 - Open source vs. proprietary
 - Subscription vs. one-time purchase vs. perpetual
 - Product keys and serial numbers
 - Software sources
 - Researching and validating legitimate sources
 - Original equipment manufacturer
 - (OEM) websites vs. third-party websites
 - Application stores
 - Removal of software
 - Unwanted
 - Unnecessary
 - Malicious
 - Software piracy

Lesson Background

Background Information:

Applications are critical to device functionality. Because of how software and applications affect day-to-day work, securing devices involves more than just firewalls and antivirus tools. It also includes making responsible software decisions. Software licensing affects user rights and device compliance. Understanding the difference between Original Equipment Manufacturer (OEM) and third-party software sources helps avoid unnecessary programs or malware. Pirated or unvetted software can compromise devices and lead to legal issues.

Materials Needed:

Materials per Class	• Lesson Guide 6.2.3 - Device Use Best Practices • Lesson Slides 6.2.3 - Device Use Best Practices • Activity Slides 6.2.3 – Best Practices, Best Choices
Materials per Student	• Student Handout 6.2.3 - Device Use Best Practices • Student Notes 6.2.3 - Device Use Best Practices • Assessment 6.2.3 - Device Use Best Practices (optional)

Lesson Background

Cyber Terms:

Term	Definition
open source	the source code of a digital product is available and can be used, modified, and distributed by anyone
proprietary	the source code of a digital product is not openly available, and users must follow the license terms from the creator of the digital product
subscription	licensing of a digital product that requires recurring payments and often provides constant updates and support
one-time purchase	licensing of a digital product using a flat fee to use the digital product, often with fewer updates and support, with newer versions requiring another purchase
perpetual license	a one-time fee, often the more expensive option, for a digital product, but allows for indefinite use often with updates and support for future services
product key	used to activate or authenticate that a software license was purchased
serial number	identification numbers reserved for physical devices, hardware, or equipment to verify the product, register for warranties or service, or used for tracking purposes
original equipment manufacturer (OEM)	the initial source of a product that can regularly ensure compatibility and security since the product has only passed through that entity
third-party retailer	distribute software or services that were manufactured by another entity, which may or may not be authorized
application stores	virtual stores that allow downloading of software or applications, which are often vetted for compatibility and security, for devices
bloatware	software that is often added by vendors that can use the device's resources without providing a benefit for the user, often considered unwanted or unnecessary
malware	any type of software that is designed to damage or disrupt a computer system, often without the permission of the user
software piracy	illegitimate copy of a software program, operating system, or other digital file without the consent of the company who manufactured it

6.2.3 - Device Use Best Practices

Guiding Question: How can we make smart software choices that protect devices and maintain legality and security for our devices?

Lesson Launch (3-5 minutes)

- Start by presenting students with the following scenario:
 - You missed seeing a movie you really wanted to see in theaters. Good news is the digital version was just released. How would you get access to this movie?
- You can click to bring up examples, but the primary takeaway that leads into the lesson is how do students know they are getting what they really want when downloading/installing movies, software, and more.

Licensing (15-20 minutes)

Open Source vs. Proprietary

- Explain the differences between open source vs. proprietary products. **Open source** means the source code is readily available to users who can use, modify, and distribute if it follows the open-source license. **Proprietary** means the source code is not available to the users, and they must follow the manufacturer's license.
- Note that proprietary does not mean it has to be paid for, and open source does not mean it is always free

Types of Paid Services

- Subscription** services are based on recurring payments and often include updates and services for as long as the subscription lasts.
- One-time** purchases offer a flat fee for using the software or service but have limited updates and support. Older versions of Microsoft Office and some operating systems are like this.
- Perpetual** licensing usually refers to a one-time fee but grants the user indefinite use and possibly access to updates and future services. This is often specialty software or software that a user knows they will use for an extended time to justify the higher cost.

Product Keys and Serial Numbers

- Product keys** are often used to activate or authenticate that a software license was purchased. An example students may understand immediately is when they purchase a digital copy of a game from a physical store and have to enter a code to download and install the game on their system.
- Serial numbers** are numbers typically reserved for physical devices, hardware, or equipment to verify the product, register for warranties or service, or used for tracking purposes. These tend to be physically marked or labelled on the equipment.

Avoiding Sketchy Software Sources (20-25 minutes)

Sourcing Software

- Note the following when sourcing software:
 - Check for legitimate sources such as the official vendor, original manufacturer, or an approved vendor.
 - Approved vendors are often licensed by the original manufacturer to sell or provide services. This license is often available in the store or on the site.
 - Authorized retailers usually advertise themselves as such and can be verified. An example includes Apple Authorized Retailer for electronics stores.
 - Do not trust third parties from ads, pop-ups, or questionable links.
 - Be cautious regarding refurbished products and software.

Sources of Hardware and Software

- **Original equipment manufacturers (OEM)** are the first entity that creates or builds a product. They are reliable and can ensure compatibility and security.
- **Third-party retailers** distribute software or services that was manufactured by another entity. This can be a good way to provide additional places to provide services of the OEM; however, some may install unneeded features or add-ons.
- **Application stores** serve as a place to download software for specific devices. They tend to vet apps before they allow it to be published in the store, ensuring compatibility and security for devices. It is important to note that these apps and devices often rely on permissions set by the user to further security.
 - Note there are unofficial application stores for various devices, which are created to provide more leeway on what a user can install. However, it is important to note that these unofficial app stores may not be compatible and could pose security risks.

Removing Software

- Unwanted or unnecessary software does not help or serve a purpose for the user. Often this is called **bloatware**, which can use up the device or system's resources.
- **Malware** can come in many forms like spyware, adware, or worse. This type of software often performs actions or makes changes without the permission of the user.
- Regardless of the software needing to be removed, it is important to use the native system tools and software designed to scan and remove all traces of the program to prevent residual files from being retained on the device or system.

Piracy

- **Software piracy** involves using an illegitimate copy of a software program, operating system, or other digital file without the consent of the company who manufactured it.
- There are several issues with piracy including:
 - Laws and regulations can often lead to fines or other punishments.

- Malicious risks are associated with pirated software due to where and how the software was downloaded.
- Pirated software may not support devices or lack crucial updates and patches making it unsecure.
- There is also the ethical side of using pirated software without the manufacturer getting credit or pay for the use of their work.

Activity: Best Practices, Best Choices (15-20 minutes)

- Display Activity Slides 6.2.3 – Best Practices, Best Choices, have students record their answers in the handout.
- Students will work in groups to examine six scenarios that involve poor choices in software use practices.
- Students will need to identify the risks associated with this choice and what could be performed to remedy the situation.
- Groups can work through the scenarios one at a time as a class or be grouped to analyze a single scenario that can be shared with the other groups afterwards.
- The scenarios:
 - Pirated Software
 - A small graphic design studio wanted to save money on Adobe Creative Cloud licenses. They downloaded illegitimate copies from a torrent site.

Risks could include software containing altered or malicious files. It may not be compatible with the studio's devices. They could also face legal issues with using illegitimate copies of a paid service. It is also unethical as a business practice. The studio should only use legitimate sources for licenses such as from the OEM or an approved vendor.
 - Third-Party Software
 - A mid-sized nonprofit downloaded a free accounting tool from an unknown third-party website instead of the official vendor site.

Risks could include software containing altered or malicious files. It may not be compatible with the organization's devices. The organization should only use legitimate sources for licenses such as from the OEM or an approved vendor. Since this is accounting software, they may want to use proprietary software or at least a product that has ongoing support and updates/ patches to prevent security flaws in the future.
 - Unlicensed Product Keys
 - A large manufacturing corporation purchased a batch of cheap Microsoft Office product keys from an online reseller.

A risk could be that the keys have already been used or were not real keys to begin with.

Improperly licensed products could lead to limited features or no functionality at all. They could also lack support, updates, and patches. The corporation should only use real product keys such as from the OEM or an approved vendor.

- Unnecessary Software

- A small medical clinic bought new laptops but never removed pre-installed trial software and games that came from the manufacturer.

The main risk is that the unnecessary software could be using up resources on devices, such as memory or storage, as seen with bloatware. The devices could also contain adware in attempts to get customers to purchase other programs that they do not need. Removal of any unnecessary software should be performed prior to distributing the laptops to employees.

- Updates?

- An e-commerce company uses legitimate software and licensing; however, they were notified by their OEM that a software update was critical for one of their tools. The IT team decided not to act since “it was still working fine.”

The risk is that an update or patch often contains security updates that close vulnerabilities or other issues with the product. Mitigation would include making sure all software was updated to the latest version to ensure that these security updates have been applied.

- Forgotten Bill

- A large hospital subscribed to cloud-based collaboration software. They forgot to renew their subscription due to a mismanaged budget.

Since the software uses a subscription-based billing plan, the hospital not only has a risk of losing access to the software, but also to any files they have created in the past. They could ensure the bill is paid, but also since this is a large hospital, they may want to look at products that offer perpetual licensing.

Putting It All Together (3-5 minutes)

- Why is it important to research software sources before downloading?

Unapproved sources could alter the product to have something malicious added to it, be unable to provide updates and patches when needed, or lack overall support.

- What are the key differences between open source and proprietary software?

Open source means the source code is available to the user, while proprietary hides the source code from the user and is more restrictive with the license.

- Why should unnecessary or unused software be removed from a device?

It can be malicious in nature or use up resources for the user that could be used elsewhere.